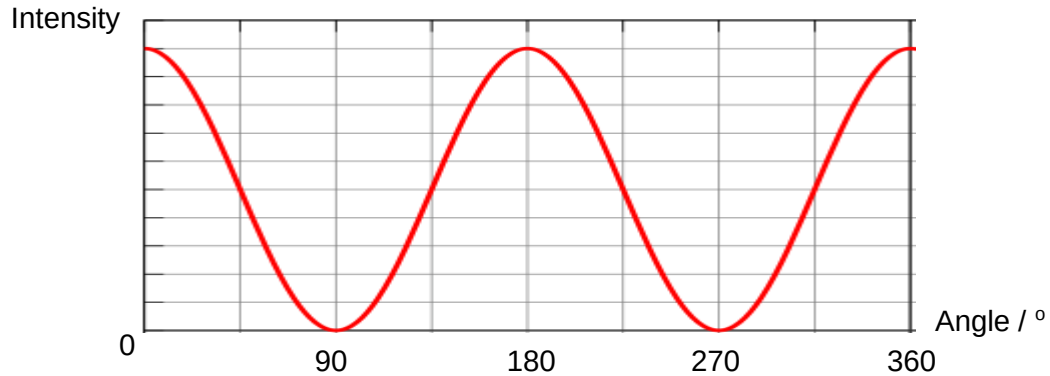


3 (a) (i)



Sinusoidally shaped graph [1]

Maximum intensity value at 0°, 180° and 360° and zero intensity at 90° and 270°. [1]

(ii) Using Malus's Law, intensity of transmitted beam, $I_t = I_o \cos^2 \theta$ [1]

Therefore angle, $\theta = \cos^{-1} \left(\sqrt{\frac{I_t}{I_o}} \right) = \cos^{-1} \left(\sqrt{\frac{4.2}{7.6}} \right) = 42^\circ$ [1]

(b) Diffraction refers to the spreading or bending of plane waves when they encounter an aperture or obstacle, [1]
whose linear dimension is comparable to the wavelength of the waves. [1]

(c)(i) For diffraction grating, $d \sin \theta = n\lambda$ [1]

Therefore, line spacing, $d = \frac{n\lambda}{\sin \theta} = \frac{3(4.3 \times 10^{-7})}{\sin 68^\circ} = 1.4 \times 10^{-6} \text{ m}$ [1]

(c)(ii) For diffraction grating, $d \sin \theta = n\lambda$

Different visible wavelength, $\lambda = \frac{d \sin \theta}{n} = \frac{(1.4 \times 10^{-6})(\sin 68^\circ)}{2} = 6.5 \times 10^{-7} \text{ m}$

[1 for $n = 2$; 1 for answer]

4(a) Electric field strength at a point is electric force per unit positive charge at that